

Principle of Inclusion & Exclusion

Examples and Propositions

$$\sim |A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Ex. Determine No. of Integers that are divisible by 2, 3, 5 or 7 between 1 and 250.

$|A|$ = No. of Integers divisible by 2

$|B|$ = " " 3

$|C|$ = " " 5

$|D|$ = " " 7

$$\sim |A \cup B \cup C \cup D| =$$

$\left\lfloor \frac{P}{Q} \right\rfloor$

$$- |A| = \left\lfloor \frac{250}{2} \right\rfloor = 125, |B| = \left\lfloor \frac{250}{3} \right\rfloor$$

$$|C| = \left\lfloor \frac{250}{5} \right\rfloor, |D| = \left\lfloor \frac{250}{7} \right\rfloor$$

$$|A \cap B| = \left\lfloor \frac{250}{2 \times 3} \right\rfloor, |A \cap B \cap C| = \left\lfloor \frac{250}{2 \times 3 \times 5} \right\rfloor$$

$$\begin{aligned} |A \cup B \cup C \cup D| &= |A| + |B| + |C| + |D| - |A \cap B| - |A \cap C| - |A \cap D| \\ &\quad - |B \cap C| - |B \cap D| - |C \cap D| + |A \cap B \cap C| + |A \cap B \cap D| \\ &\quad + |A \cap C \cap D| + |B \cap C \cap D| - |A \cap B \cap C \cap D| \end{aligned}$$

=

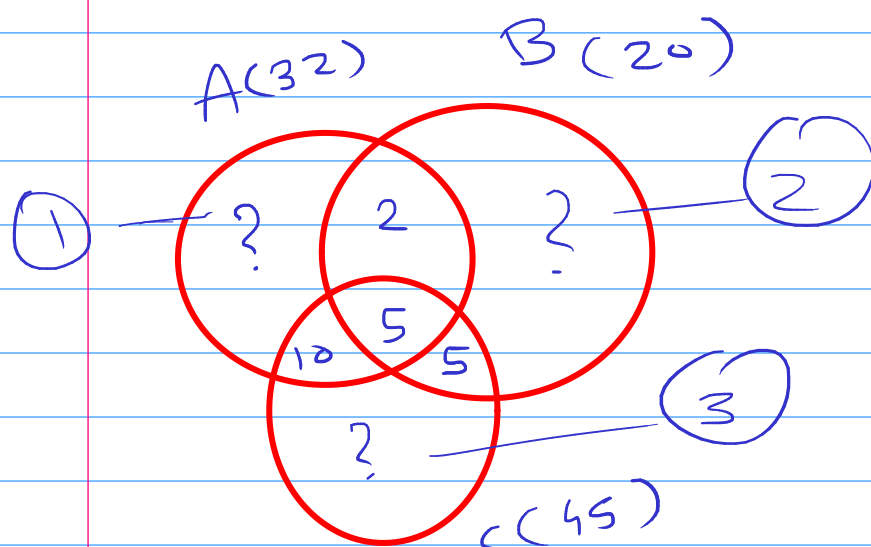
Ex: Total = 100 $|A \cap B \cap C| = ?$
 $|A| = 32$ 30 Do not study any Subject
 $|B| = 20$
 $|C| = 45$ Total = $|A \cup B \cup C| + 30$
 $|A \cap C| = 15$ $100 = |A \cup B \cup C| + 30$
 $|A \cap B| = 7$ $|A \cup B \cup C| = 70$
 $|B \cap C| = 10$

$$|A \cup B \cup C| = 32 + 20 + 45 - 15 - 7 - 10 + |A \cap B \cap C|$$

$$70 =$$

$$|A \cap B \cap C| = 5 \quad \text{--- (i)}$$

~ Students Studying only Maths
 " " " Physics
 " " " Biology



Propositions

- ~ Declarative Statement which can be either entire True or entire False.
- ~ It is 4 o'clock - ✓ (Valid Proposition)
- ~ What is current time? (Not a Proposition)
- ~ Tautology: Proposition which is always TRUE
"Sun rises in east"
- ~ Contradiction: Proposition which is always FALSE.
"Sun sets in east"

* Logical Connectives

1. Conjunction ($\wedge$)

P	Q	$(P \wedge Q)$
T	T	T
T	F	F
F	T	F
F	F	F

P = I go to Mumbai
Q = I go to Pune.

2. Disjunction ($\vee$)

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

3. Negation

P	$\neg P$
T	F
F	T

4. X-OR ($\oplus$)

P	Q	$P \oplus Q$
T	T	F
T	F	T
F	T	T
F	F	F

5. Conditional ($P \rightarrow Q$) (If P, then Q)

*	P	Q	$P \rightarrow Q$
	<u>T</u>	T	T
	<u>T</u>	F	F
$\rightarrow$	F	T	T ✓
	F	F	T ✓

$\sim P = I$ get 10/10 SPI

$Q = I$ get i-Phone as gift

$P \rightarrow Q = \text{Promise}$

* Bi-Conditional ($P \leftrightarrow Q$) (P if and only if Q)

P	Q	$P \leftrightarrow Q$
T	T	<u>T</u>
T	F	F
$\sim \rightarrow$ F	T	F
F	F	<u>T</u>

$\Rightarrow$ P is necessary
and sufficient for Q.

* Prepare Logic table for

$$\Rightarrow ((P \wedge Q) \vee (\bar{P} \wedge \bar{Q})) \rightarrow P$$

If (x) then y
 $x \rightarrow y$

P	Q	
T	T	T
T	F	T
F	T	T
F	F	F